

Online Appendix to

Weaponized Reputation:

Experimental Evidence on the Misuse of Coordinated Punishments

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B. Experimental Instructions for the Public Commitment Treatment

The instructions below are translated from the original Chinese version and correspond to the public-commitment treatment. In the silent-commitment treatment, Step 2 is identical except that Player B does not observe Player A's evaluation rule before choosing the payment amount.

Welcome to this economics experiment. In this experiment, your payoff will depend on your decisions, the decisions of other participants, and some random probabilities.

Basic Rules:

- This is an economics experiment. All rules and random probabilities mentioned in these instructions will be executed strictly as described; there is no deception.
- All your decisions will be completely anonymous. Other participants will not know your true identity.
- During the experiment, please remain quiet. Communicating with other participants or using electronic devices such as mobile phones or earphones is strictly prohibited. If you have any questions, please raise your hand, and an experimenter will come to your seat to assist you.
- At the end of the experiment, your earnings will be converted according to the rules and paid to you in **cash**.

This experiment consists of **two tasks** (Task 1 and Task 2). Your total payoff will be determined jointly by your earnings from these two tasks. Throughout the experiment, all payoffs will be calculated in **Experimental Currency Units (ECU)**.

Task 1

In Task 1, all participants in this room will be randomly divided into two roles in equal proportions: **Player A** and **Player B**. Once assigned, your role will remain unchanged throughout the entire experiment.

Task 1 will consist of **5 supergames** in total. Each “**supergame**” contains a certain number of “**rounds**”, and the game content in each “**round**” is the same. **Every round played in Task 1 will count toward your final payment.**

Decision Process for Each “Round”

In each round, the system will automatically pair one Player A with one Player B. **At the beginning of each round, both Player A and Player B will receive an initial endowment of 40 ECU.** The interaction in each round consists of the following three steps:

Step 1: Player A’s Decision and Setting the Evaluation Rule

First, Player A needs to choose an “**effort level**” (any integer from 0 to 12, inclusive). A higher effort level will bring more payoff to Player B, but Player A will also incur increasing marginal costs. The specific relationship between costs and payoffs is shown in the table below.

TABLE B1. Effort cost schedule used in the experiment instructions.

Player A’s Effort Level	0	1	2	3	4	5	6	7	8	9	10	11	12
Cost Incurred by Player A	0	1	3	6	10	15	21	28	36	45	55	66	78
Payoff Generated for Player B	0	8	16	24	32	40	48	56	64	72	80	88	96

At the same time, Player A must pre-set an “evaluation rule,” namely a “positive review threshold.” Regarding the payment amount Player B will subsequently choose to make, Player A must set the following rule while choosing the effort level: If the amount paid by Player B is equal to or greater than the “**positive review threshold**,” the system will automatically give your opponent a “**positive review**.” If the amount paid by Player B is less than the “**positive review threshold**,” the system will automatically give your opponent a “**negative review**.”

Step 2: Player B’s Decision

Player B will observe the effort level chosen by Player A in the current round and the resulting payoff points, **and can fully see the “evaluation rule” (i.e., the “positive review threshold”) set by Player A in Step 1.** Subsequently, Player B needs to decide how much “payment amount” to pay to Player A.

- Player B can choose to pay any **non-negative integer** amount of points from **0 to 100**, inclusive.

Step 3: Player A's Evaluation of Player B

Based on the actual payment amount given by Player B, and comparing it against the “**evaluation rule**” (i.e., the “**positive review threshold**”) set by Player A in Step 1, the system will **automatically** give Player B a “**positive review**” or a “**negative review**.”

- **Public Reputation Board:** The “positive review” or “negative review” Player B receives in this round will be recorded on the public reputation board for the current supergame. In every subsequent round of the current supergame, **future Player As matched with this Player B will be able to see the sequence record of “positive/negative reviews” this Player B received in all previous rounds of this supergame, not just the total number of “positive” and “negative” reviews.**

** Note: Player A can only see the evaluations received by their matched opponent, Player B, in all previous rounds of the current supergame. They cannot see the effort levels chosen by Player B's opponents in each round of this supergame, nor the payment amounts Player B chose to give their opponents. Conversely, Player B cannot see the information regarding the effort level, positive review threshold, or evaluations given to opponents by their matched opponent, Player A, in previous rounds of the current supergame.*

Single-Round Payoff Calculation for Task 1

After the three steps of this round are completed, the single-round payoffs for both parties are calculated as follows:

- **Player A's Payoff for the Current Round** = 40 (Initial Endowment) - Cost corresponding to the effort level + Payment amount given by Player B
- **Player B's Payoff for the Current Round** = 40 (Initial Endowment) + Payoff points generated by Player A's effort - Payment amount given to Player A

Matching and “Supergame” Termination Rules

After each round ends, the system will determine how the game continues based on the following rules:

- **Supergame Termination and Number of Rounds:** After each “round” ends, the computer will generate a random number for the entire room to determine whether the current “supergame” will proceed:
 - There is an **80% probability** that the current supergame will proceed to the next round.
 - There is a **20% probability** that the current supergame will end immediately.
- **Matching Rules for Different Rounds within Each “Supergame”:** If the current supergame proceeds to the next round, all participants in this room will be re-matched. **The system will ensure that the opponent you meet in each round is different from the one in the previous round.**

- **Entering a New “Supergame”:** When a supergame ends (due to triggering the 20% probability), everyone will jointly enter the next supergame (a total of 5 supergames will be played). **At the beginning of a new supergame, all previous reputation (positive/negative review) records for Player B will be cleared, but your role remains unchanged.**
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Task 2: Bomb Risk Game

Task 2 consists of only **one round** of decision-making.

Task Rules:

100 boxes will be displayed on the screen. Among these 100 boxes, 99 are empty, and only 1 box contains a “bomb.” The computer will randomly generate a number between 1 and 100 as the number of the box containing the bomb. You do not know which box the bomb is hidden in.

Your Decision:

You need to decide how many boxes you want to collect. You can enter any integer between **0 and 100**.

Task 2 Payoff Calculation:

Your payoff depends on the number of boxes you collect and the bomb number randomly generated by the computer:

- **Success (Bomb not collected):** If the randomly generated bomb number is **greater than** the number of boxes you chose to collect (meaning the bomb is not in the boxes you collected), you will receive a reward. For every box collected, you will earn **1 ECU**.
- **Failure (Bomb collected):** If the randomly generated bomb number is **less than or equal to** the number of boxes you chose to collect (meaning you collected the box containing the bomb along with the others), your Task 2 payoff will become **0 points**.

Final Payment

In the final stage of the experiment, the system will calculate your cash remuneration as follows:

- Task 1 Payoff:** The system will add the points you earned in **every round** of all 5 supergames.
(Task 1 Calculation Formula: *Sum of points from all Task 1 rounds = Final points for Task 1*)
- Task 2 Payoff:** This is the experimental points you actually earned in the Bomb Risk Game.

The exchange rate is **16 ECU = 1 RMB**. Your earnings from the experimental tasks are calculated as follows:

Total cash payment in RMB = (Final points for Task 1 + Payoff points for Task 2) ÷ 16

At the end of the experiment, there will be a brief questionnaire (your information will be kept confidential).

If you have understood the above instructions, please wait quietly; the experiment is about to begin. If you have any questions, please raise your hand now.

Now, you need to finish a comprehension quiz to ensure you have understood the instructions. You can only proceed to the experiment after answering all quiz questions correctly.

C. Comprehension Quiz

The quiz below is translated from the original Chinese version. The displayed questions were identical in the two treatments. The responses treated as correct for Question 5 and the second part of Question 10 reflected whether Player B could observe Player A's positive review threshold in the relevant treatment.

Comprehension Quiz (All Answers Must Be Correct)

Please answer the following 13 questions. Questions 1–11 concern Task 1, and Questions 12–13 concern Task 2. If any answer is incorrect, the system will identify the relevant question number, and you may try again.

- 1. Multiple choice.** Which statement about your assigned role (“Player A” or “Player B”) is correct?
 - A. Roles are reassigned in every round.
 - B. Roles are reassigned at the beginning of every new supergame.
 - C. Roles remain fixed throughout the experiment.
 - D. The two roles alternate.

- 2. Multiple choice.** Suppose the experiment is currently in Round 5 of Supergame 2. Which statement about the matching procedure is correct?
 - A. Participants are randomly matched across the room, so you may be matched with the same opponent as in Round 4 of Supergame 2.
 - B. Participants are randomly matched across the room, but the program ensures that your opponent differs from your opponent in Round 4 of Supergame 2.
 - C. You always interact with a fixed opponent, so you continue to play with the same opponent as in Round 4 of Supergame 2.
 - D. Participants wait in a queue, and those who finish Round 4 of Supergame 2 first are randomly matched with one another first.

- 3. Fill in the blanks.**
 - a. Suppose the experiment is currently in Round 5 of Supergame 2. After this round is completed, the probability that the next round is Round 6 of Supergame 2 is _____ percent.
 - b. Suppose the experiment is currently in Round 8 of Supergame 3. After this round is completed, the probability that the next round is Round 1 of Supergame 4 is _____ percent.

4. Multiple choice. Suppose the experiment is currently in Round 5 of Supergame 2. When choosing the effort level and positive review threshold, which information can Player A see about the Player B matched with them in the current round?

- A. Player B's payments to opponents and the positive/negative reviews Player B received in every round of Supergame 1 and the first four rounds of Supergame 2.
- B. The positive/negative reviews Player B received in every round of Supergame 1 and the first four rounds of Supergame 2.
- C. The effort levels and positive review thresholds chosen by Player B's matched opponents in every round of Supergame 1 and the first four rounds of Supergame 2.
- D. Player B's payments to opponents and the positive/negative reviews Player B received in the first four rounds of Supergame 2.
- E. The positive/negative reviews Player B received in the first four rounds of Supergame 2.
- F. The effort levels and positive review thresholds chosen by Player B's matched opponents in the first four rounds of Supergame 2.
- G. The payment Player B chooses in Round 5 of Supergame 2.
- H. Player A cannot see any information about the matched Player B when making this decision.

5. Multiple choice. Suppose the experiment is currently in Round 5 of Supergame 2. When choosing how much to pay the Player A matched with them in the current round, which information can Player B see about Player A?

- A. Player A's effort levels and positive review thresholds in every round of Supergame 1 and the first four rounds of Supergame 2.
- B. The positive/negative reviews Player A gave to matched opponents in every round of Supergame 1 and the first four rounds of Supergame 2.
- C. Player A's effort levels and positive review thresholds in the first four rounds of Supergame 2.
- D. The positive/negative reviews Player A gave to matched opponents in the first four rounds of Supergame 2.
- E. Player A's effort level and positive review threshold in Round 5 of Supergame 2.
- F. Player A's effort level in Round 5 of Supergame 2.
- G. The positive/negative review Player A gives in Round 5 of Supergame 2.
- H. Player B cannot see any information about the matched Player A when making this decision.

- 6. Fill in the blank.** At the beginning of each round, before Player A or Player B makes any decision, each participant automatically receives _____ ECU.
- 7. Fill in the blanks.** Player A chooses an effort level of 3. Player A incurs a cost of _____ ECU, and this effort level generates a benefit of _____ ECU for Player B.
- 8. Fill in the blanks.** Player A chooses an effort level of 5 and sets the positive review threshold at 30 ECU. Player B chooses to pay Player A 25 ECU. Player A's payoff in this round is _____ ECU, Player B's payoff is _____ ECU, and Player B receives a _____ (positive/negative) review.
- 9. Fill in the blanks.** Player A chooses an effort level of 7 and sets the positive review threshold at 40 ECU. Player B chooses to pay Player A 42 ECU. Player A's payoff in this round is _____ ECU, Player B's payoff is _____ ECU, and Player B receives a _____ (positive/negative) review.
- 10. Fill in the blanks.** Regarding the information available to Player B at the end of a round:
- Player B _____ (can/cannot) learn the review received in that round.
 - Player B _____ (can/cannot) learn the positive review threshold set for that round by the matched Player A.
- 11. Multiple choice.** How is the final cash payment from Task 1 calculated?
- One round is randomly selected from one randomly selected supergame.
 - One round is randomly selected from the supergame with the highest total payoff.
 - One supergame is randomly selected; the points from all its rounds are summed and divided by the number of rounds to obtain the average points per round in that supergame.
 - The points earned in every round of all five supergames are summed.
- 12. Fill in the blank.** In Task 2, suppose you choose to collect the first 40 boxes and the bomb is randomly placed in Box 73. Your payoff from Task 2 is _____ ECU.
- 13. Fill in the blank.** In Task 2, suppose you choose to collect the first 60 boxes and the bomb is randomly placed in Box 59. Your payoff from Task 2 is _____ ECU.